When Do Attention-Head Ablations Support Causal Claims? Projection-Level Confounds, Floor Effects, and Matched Controls

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Abstract

Attention-head ablation, replacing a head's output and measuring the resulting change in task performance, is a standard way to infer which components of a language model are causally responsible for a behavior. Using GPT-2 small as a case study, with replications on DistilGPT2 and Qwen2.5-1.5B-Instruct, we show that this recipe is fragile in ways that are rarely checked. First, a natural implementation of "zero out head h" (overwriting a channel slice of the attention block's *returned* output) does not ablate a single head, because the output projection has already mixed all heads. On a controlled induction-style copying task, this naive method is essentially uncorrelated with a corrected, pre-projection ablation (Pearson $r = 0.057$) and selects a disjoint top-5 set of heads. The confound recurs on both other models. Second, accuracy cannot register further degradation at a floor (0.0% on an off-target arithmetic task) and can miss confidence changes near a ceiling (99.5% on copying), while log-probability stays graded. Third, with a discovery/held-out split and 1,000 matched random-head and layer-matched control draws, the top-5 selected heads' held-out effect exceeds both controls (Monte Carlo $p = 0.001$), and it does so in all 20 random re-partitions. That effect, however, depends on the replacement value: the same GPT-2 heads' effect falls from 0.412 nats to 0.005 under mean-ablation and to 0.020 under resample-ablation, and on indirect object identification zero-ablation shares only 2 of its top-5 heads with mean or resample ablation. Task specificity is not established on GPT-2: the contrast with arithmetic is inconclusive ($p = 0.078$; $p = 0.274$ at $N = 500$), and against a format-matched sequence-continuation control it disappears ($p = 0.567$). A single-head ablation result is not self-certifying. It requires verified projection semantics, a continuous metric away from floor and ceiling, held-out matched controls, and a stated replacement value checked against an on-distribution baseline.

1 Introduction

Interpretability researchers frequently ablate an individual attention head, replacing its contribution and re-measuring task performance, and interpret a resulting drop as evidence that the head is causally responsible for some computation. Whether this inference is valid depends on choices that are rarely audited: what, mechanically, gets replaced relative to the attention block's linear mixing structure; what it is replaced with; whether the evaluation metric has room to move at all; and whether the identified heads are validated against an appropriate null rather than just reported as the top of a ranking. This paper asks a narrow, mechanical question: **under what conditions does single-head**

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: Interpretability as a Science.

ablation support a defensible causal interpretation, in the sense of an effect that exceeds matched controls on held-out data?

We answer it by auditing and fixing a head-ablation implementation, designing a controlled induction-style copying task with behavioral headroom alongside an off-target floor task and a format-matched control task, and running a discovery/held-out pipeline with random and layer-matched controls. The primary model is GPT-2 small (124M parameters, 12 layers $\times$ 12 heads); DistilGPT2 and Qwen2.5-1.5B-Instruct serve as replications, and indirect object identification (IOI) on GPT-2 as an established circuit task. We report four findings, one per research question (RQ0–RQ3), matching the subsections of Section 4.

(1, RQ0) The ablation implementation matters. A hook registered on the whole attention module's output, a natural way to write "zero out head h", zeroes channels *after* the block's output projection has already mixed all heads together, so it does not isolate one head's contribution. On our controlled copying task, this naive method's per-head effect vector is essentially uncorrelated with a corrected, pre-projection version (Pearson $r = 0.057$), and the two methods' top-5 "important" heads share no head. The same holds on DistilGPT2 and Qwen2.5.

(2, RQ1) Binary accuracy loses information at floor and near ceiling. GPT-2 small scores 0.0% on an off-target arithmetic task and 99.5% on our on-target copying task. At the floor, accuracy cannot register further degradation. Near the ceiling, it can register a sufficiently large drop but may hide confidence changes that leave the predicted token unchanged. Teacher-forced log-probability of the gold continuation remains graded in both regimes.

(3, RQ2) With proper controls, five heads show a held-out-replicated intervention effect that exceeds random and layer-matched baselines, but their task specificity is not established on GPT-2. The per-head ranking is stable across splits (Spearman $\rho = 0.974$ over all 144 heads), and the top-5 discovery-selected heads' mean held-out effect clears both control distributions (Monte Carlo $p = 0.001$ each), in the original partition and in all 20 random re-partitions. Their contrast against an off-target arithmetic task is inconclusive ($p = 0.078$; $p = 0.274$ in a pre-committed $N = 500$ rerun), and against a format-matched sequence-continuation task it disappears ($p = 0.567$).

(4, RQ3) The replacement value changes which heads are selected. Under mean-ablation and under resample-ablation, the five zero-selected GPT-2 heads have almost no held-out effect (0.005 and 0.020 vs. 0.412 nats), and reselection yields sets that share no head with them. On IOI, zero-ablation shares only 2 of its top-5 heads with mean or resample ablation. Baseline dependence is partial on DistilGPT2 and small on Qwen2.5, so it must be checked per model.

Our contribution is a worked, fully reproducible example (code, seeds, and every number traceable to a saved result file) showing that a positive *or* null single-head ablation result is not self-certifying: a causal interpretation additionally requires (i) verifying the intervention's projection semantics, (ii) a continuous metric that is not saturated, (iii) matched controls evaluated on data not used for head selection, and (iv) a stated replacement value checked against at least one on-distribution baseline.

2 Why naive head ablation can mislead

GPT-2's attention block computes a per-head output for each head, concatenates the per-head outputs, and applies a single dense output projection (W_O, `c_proj` in the standard `transformers` implementation) before adding the result to the residual stream. Elhage et al. [1] describe W_O as half of each head's output-value (OV) circuit. Because W_O spans the concatenated head dimension, each output channel can receive contributions from all heads. A hook that zeroes a contiguous, head-sized slice of the block's *returned* output (registered as a forward hook on the whole attention module) therefore does not remove one head's contribution. It removes an arbitrary linear combination of all heads' contributions to a subset of output channels, after mixing has already happened. The correct intervention instead zeroes the corresponding slice of W_O's *input*, i.e., the concatenated, not-yet-mixed per-head values, which requires a pre-hook on the projection itself rather than a post-hook on the block.

We did not construct this as a strawman: it is the hook used by this project's own original head-ablation implementation, which we found when we audited it (preserved, unused for any reported result, as `src/legacy_invalid_ablation.py`). We use it as the "naive" method in our RQ0

comparison (Section 4). The methods use the same nominal head index, model, and input. Their activation interventions differ only in where they are applied within the block.

Scope of the failure. The failure concerns an implementation that treats a contiguous slice of the already-projected block output as if it were one head. The confound is a property of where the intervention is placed, not of a particular library. TransformerLens [2] exposes a per-head, pre-projection hook (`hook_z`), which corresponds to our valid intervention, but it also exposes the block's mixed output (`hook_attn_out`), where slicing channels would reproduce the naive method. NNsight [3] exposes both the output of an attention module and the input of its output projection, so the same choice arises there. Our results do not show that these libraries implement the error, and we did not measure how often the naive placement occurs in published work. They show that the semantics of an intervention must be checked against the model's computational graph.

Choice of replacement value. Zero-ablation sets a head's output to a value it may never take during normal operation. Circuit analyses therefore often replace activations with dataset means [4], and Li and Janson [5] propose optimal ablation as an alternative. We use zero-ablation for the primary analysis because it is a common default, and we test whether the conclusions survive mean and resample replacement in Sections 4.4–4.5.

Separately, prior work has repeatedly cautioned that single-component interventions and attention weights are easy to over-interpret without controls: Jain and Wallace [6] show attention weights alone are often uncorrelated with feature importance, and Michel et al. [7] show many heads can be pruned post hoc without hurting overall performance, i.e., that a head being prunable does not mean it was doing nothing task-specific. More recent circuit-style work explicitly validates causal claims about specific components against matched baselines and held-out data [8, 4, 9]. Our contribution is narrower and more mechanical: we isolate the projection-level implementation confound specifically, and combine it with a floor/ceiling-aware metric and a minimal control battery in one end-to-end worked example.

3 Experimental design

Models. The primary model is GPT-2 small (124M parameters, 12 layers, 12 heads/layer, 144 heads total), the standard Hugging Face `transformers` implementation (v4.57.3, torch 2.8.0), run on CPU. Replications use DistilGPT2 (6 layers, 72 heads), which is distilled from GPT-2 and therefore not an independent architecture, and Qwen2.5-1.5B-Instruct [10] (28 layers, 12 query heads per layer, 336 heads), whose output projection is hooked in the same pre-projection position.

Tasks. *Controlled induction-style copying* (target task, expected headroom): synthetic four-word sequences of the form "$w_1\ w_2\ w_3\ w_4\ .\ w_1\ w_2\ w_3$" $\rightarrow$ "$\ w_4$", with $w_1, \ldots, w_4$ drawn without replacement from a pool of common English words verified to tokenize as exactly one byte-pair-encoding (BPE) token, a minimal instance of the induction-head phenomenon described by Olsson et al. [11] and traced to two-layer head composition by Elhage et al. [1]. We generated $N = 200$ examples (seed 0) and split them 50/50 into a *discovery* set (used only to select candidate heads) and a *held-out* set (used only to evaluate them), fixed before any head effect was computed. *Arithmetic* (off-target control task): "Q: a+b=" $\rightarrow$ "$\ a$+b", a, b drawn uniformly from $[1, 20]$, $N = 200$ (seed 1), used only to test task-specificity of already-selected heads, never for selection. *Ordered-sequence continuation* (format-matched control task, added for the camera-ready version): "$x_1\ x_2\ x_3\ x_4\ .\ x_5\ x_6\ x_7$" $\rightarrow$ "$\ x_8$", where $x_1, \ldots, x_8$ are eight consecutive items of a familiar ordering (numerals 1–99, months, capital letters, number words one–twenty). Every non-wrapping window is used once, giving $N = 129$ examples (92 numerals, 19 letters, 13 number words, 5 months). The template, the period, and the single-token, space-prefixed answer match the copying task, but the answer never appears in the prompt, so it cannot be copied in context. It is predicted from knowledge of the ordering, a behavior attributed to "successor heads" [12]. Every generated example is checked so that the question's tokenization is an exact token-level prefix of question+answer (BPE merges are not always prefix-stable across concatenation). Examples failing this check are dropped (and, for the two sampled tasks, regenerated).

Choice of tasks. The tasks are deliberately simple. We use them to isolate methodological issues in head ablation, not to benchmark language understanding. Copying gives a predictable continuation with behavioral headroom. Arithmetic is an off-target diagnostic that also exposes accuracy at a floor, and it is not a matched control: it differs from copying in template, computation, answer

tokenization, and baseline performance. The sequence task is a matched control in template, answer length, and baseline (Section 4.3). Simple tasks make the diagnostic results interpretable, but they limit conclusions about more complex behaviors, so we also test an established circuit task, indirect object identification (Section 4.5).

Metrics. *Teacher-forced exact match*: the greedy argmax token at every answer position must equal the gold token (binary). *Teacher-forced log-probability*: $\sum_t \log p(\text{gold token}_t \mid \text{prefix})$ over the answer's tokens (continuous, natural log). Both require a single forward pass per batch (no autoregressive generation).

Ablation methods. `naive_post_projection`: a forward hook on the attention module's output, zeroing a head-sized channel slice of the returned (already-projected) tensor. This is invalid and used only for the RQ0 comparison. `head_ablation`: a forward pre-hook on the output projection's input, zeroing the corresponding pre-mix slice. Every causal claim uses this pre-projection location. The primary analysis replaces the slice with zero. The robustness checks replace it with the slice's mean (*mean-ablation*; computed over all non-padding positions of the discovery split only and applied unchanged to every split, so held-out data never inform the intervention) or with its activation on another input (*resample-ablation*). For copying, the source is a different example from the same split whose words split into the same numbers of tokens, so every position plays the same role (one discovery example has no such partner and is dropped, leaving 99); for IOI, it is a corrupted prompt (Section 4.5). These semantics are asserted by executable tests that run before the pipeline and ship with the code: on a small GPT-2 configuration they check that the pre-hook zeroes exactly the requested head's slice of the projection input and leaves every other channel unchanged, that removing the hook exactly restores the unablated output, and that the naive hook alters the logits while leaving that projection input untouched.

Head selection. For all 144 heads, we compute the mean $\Delta = \overline{\text{log-prob}}_{\text{baseline}} - \overline{\text{log-prob}}_{\text{ablated}}$ over a data split, using one batched forward pass per head (not per example). Candidate heads are the top-5 by Δ on the discovery split only.

Controls. *Random-head control*: 1,000 draws of 5 heads sampled uniformly without replacement from all 144. *Layer-matched control*: 1,000 draws that preserve the candidate set's size and layer composition while replacing every candidate identity, drawing each layer's replacements without replacement from that layer's non-candidate heads. Both exclusions matter whenever one layer supplies several candidates, as ours does: replacing each candidate independently would let a single head stand in for several candidates, and would let a candidate head enter its own control distribution. Both controls are built by resampling the already-computed 144-head Δ vector, so they cost no additional forward passes. These two controls ask whether the candidates' effect is unusual *among heads*. A separate *specificity* contrast asks whether it differs *between tasks*: the candidates' held-out copying Δ minus their Δ on a control task, compared with the same quantity for the 1,000 random head sets. Passing the head controls does not imply passing the task contrast.

Statistics. Where applicable, example-level point estimates are reported with percentile-bootstrap 95% confidence intervals (CIs; 2,000 resamples over examples). Control ranges are the 5th–95th percentiles of 1,000 draws. One-sided Monte Carlo p-values use the finite-sample correction $(k + 1)/(B + 1)$, where k of $B = 1{,}000$ control draws match or exceed the candidate statistic. Rank stability across splits is measured with Spearman correlation over all 144 heads.

Reproducibility. Seeds (induction generation 0, arithmetic generation 1, control sampling 42), package versions, and the experiment configuration are recorded with every saved result. The full pipeline (`scripts/run_neurips_experiments.py`) reruns end-to-end in 198.99 s for the primary GPT-2 configuration on one 10-core Apple M5 CPU with 24 GB unified memory and no GPU (peak resident memory 1.94 GiB), 57.62 s for DistilGPT2, and 428.56 s for GPT-2 at $N = 500$ (3.22 GiB). Each of the three configurations was then rerun independently; all three reproduced their saved result files exactly, byte for byte, apart from the recorded timestamp. The camera-ready partition and sequence-control analyses (`scripts/run_camera_ready_robustness.py`) took 301.43 s (1.90 GiB) for GPT-2 and 82.08 s (1.63 GiB) for DistilGPT2, and the copying resample analysis (`scripts/run_resample_copying.py`) took 914.99 s (2.14 GiB) and 325.98 s (2.10 GiB). The software environment, model revisions, file hashes, storage, and verification commands are recorded in `neurips/experiment_results/reproducibility_provenance.md`. Every number in this paper is read from a result JSON file, and a script (`scripts/verify_paper_numbers.py`) checks the printed values against those files.

4 Results

4.1 RQ0: Does the ablation implementation matter?

On the controlled copying-task discovery split, the naive and correct per-head Δ vectors are nearly uncorrelated (Pearson $r = 0.057$, Spearman $\rho = 0.110$; scatter plot in Appendix A), with a mean absolute per-head disagreement of 0.067 nats. The top-5 heads selected by each method share **zero** heads. The same pattern holds on DistilGPT2 ($r = 0.001$) and on Qwen2.5-1.5B-Instruct ($r = 0.108$, $\rho = 0.176$), each with 0/5 top-head overlap (Appendices B and C). The methods use the same nominal head index, model, and input, but apply the activation intervention at different points in the block (Section 2). This is not a hypothetical failure mode: the naive intervention is a natural, easy-to-write hook, and it is the one this project originally used (Section 2), not one constructed to fail.

4.2 RQ1: Metric behavior at floor and near ceiling

GPT-2 small is at a behavioral floor on the arithmetic control task (0.0% teacher-forced exact match, 95% CI $[0.0\%, 0.0\%]$, $N = 200$) and near a behavioral ceiling on the controlled induction-style copying task (99.5%, CI $[98.5\%, 100.0\%]$). At the floor, binary accuracy cannot detect further degradation. Near the ceiling, it can detect a sufficiently large damaging effect, but a substantial confidence decrease may leave every argmax prediction unchanged. Log-probability remains graded: mean log-prob is -7.44 nats (CI $[-7.53, -7.35]$) on arithmetic and -0.83 nats (CI $[-0.90, -0.76]$) on the copying task (Table 1, Figure 1A), and, as shown next, individual-head ablation moves it substantially and reproducibly. This is why we select and validate candidate heads using log-probability rather than accuracy throughout. A post hoc check of one alternative answer format shows that the GPT-2 floor is not an artifact of the leading space in the gold answer: exact match remains 0.0% when the answer is written without it. This is not true for every model. Qwen2.5-1.5B-Instruct also scores 0.0%, because its tokenizer splits " 24" into a bare space token followed by digits and the model never predicts the bare space. Without the space it scores 23.0% (Appendix C). A floor in accuracy can therefore reflect the answer format rather than missing capability. Log-probability stays graded in either case (Qwen2.5: -5.31 nats).

4.3 RQ2: Specificity of selected-head intervention effects

The top-5 discovery-selected heads are Layer 0 Head 0, Layer 0 Head 7, Layer 1 Head 3, Layer 11 Head 0, and Layer 0 Head 10 (four in the first two layers, one in the last layer). Two distinct pieces of evidence support them. First, the per-head effect ranking is highly stable across the discovery and held-out splits (Spearman $\rho = 0.974$ over all 144 heads), so head ordering is not an artifact of the discovery sample. Second, the candidates' mean held-out effect, 0.412 nats (CI $[0.353, 0.474]$), is far outside both the matched random-head control distribution (mean 0.029, 5th–95th percentile range $[-0.033, 0.113]$; Monte Carlo $p = 0.001$, 0/1,000 draws matched or exceeded the candidates) and the layer-matched control distribution (mean 0.051, 5th–95th percentile range $[0.001, 0.098]$; Monte Carlo $p = 0.001$; Figure 1B). This establishes a held-out-replicated intervention effect on the controlled copying task that exceeds what an arbitrary head, or an arbitrary head from the same layers, would produce. Whether that effect is specific to the copying task requires the separate contrast below.

Task specificity is not established (Figure 1C). The specificity statistic (held-out copying-task Δ minus arithmetic-control Δ) is 0.139 nats for the candidates, versus a random-control distribution with mean 0.011 and 5th–95th percentile range $[-0.135, 0.165]$ (Monte Carlo $p = 0.078$). A single pre-committed rerun with $N = 500$ per task preserves the held-out effect over both head controls ($p = 0.001$ each) but weakens the contrast to 0.056 nats ($p = 0.274$). We therefore treat the arithmetic contrast as inconclusive. Even a significant contrast against arithmetic would be hard to interpret, because the two tasks differ in template, computation, tokenization, and baseline performance.

A format-matched control removes the GPT-2 contrast. We therefore added the ordered-sequence task (Section 3), whose template, answer length, and baseline match copying (GPT-2: 96.9% exact match, -0.57 nats) but whose answer cannot be copied from the prompt. This analysis and the partition analysis below were specified in writing before they were run and are reported as run. Ablating the five GPT-2 candidates lowers the sequence-task log-probability by 0.447 nats, far more

Table 1: Primary GPT-2 RQ1/RQ2 quantities. All values are computed from `gpt2_results.json` (Section 3). CIs are 95% percentile bootstrap over examples except where noted. "Control range" is the 5th–95th percentile of 1,000 control draws, not a CI.

Quantity	Value	Interval	N / draws
Copying-task accuracy (ceiling)	0.995	[0.985, 1.000]	200
Arithmetic accuracy (floor)	0.000	[0.000, 0.000]	200
Copying-task log-prob (nats)	-0.83	$[-0.90, -0.76]$	200
Arithmetic log-prob (nats)	-7.44	$[-7.53, -7.35]$	200
Candidate Δ, held-out	0.412	[0.353, 0.474]	100
Random-control Δ, held-out	0.029	$[-0.033, 0.113]^{\dagger}$	1000
Layer-matched Δ, held-out	0.051	$[0.001, 0.098]^{\dagger}$	1000
Rank stability (disc. vs. held-out, ρ)	0.974	–	144 heads
Candidate specificity	0.139	–	–
Random-control specificity	0.011	$[-0.135, 0.165]^{\dagger}$	1000

Monte Carlo p(candidate $\geq$ random control, held-out Δ) $= 0.001$; p(candidate $\geq$ layer-matched, held-out Δ) $= 0.001$; p(candidate $\geq$ random control, specificity) $= 0.078$. †5th–95th percentile.

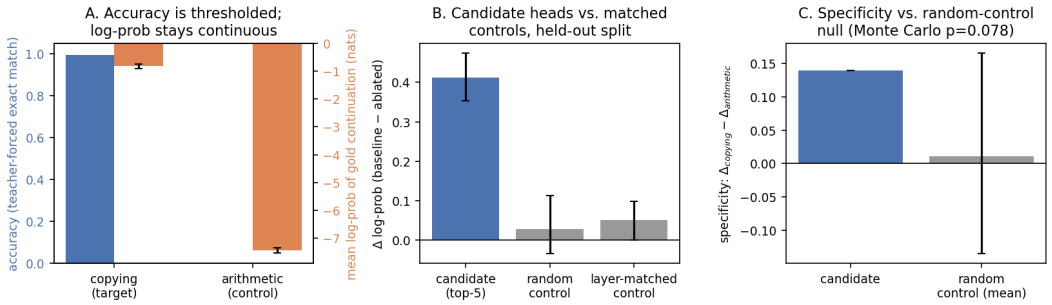


Figure 1: **A.** Accuracy cannot show further degradation at the arithmetic floor and may hide confidence changes near the copying-task ceiling. Log-probability stays continuous. **B.** The top-5 candidate heads' held-out effect far exceeds matched random-head and layer-matched control distributions (error bars: 95% CI for the candidate, 5th–95th percentile for controls). **C.** Specificity (copying minus arithmetic effect) is larger for candidates than the random-control mean, but the contrast is inconclusive (Monte Carlo $p = 0.078$; $p = 0.274$ at $N = 500$) and disappears against a format-matched control (Section 4.3).

than random head sets do (mean 0.051, $p = 0.001$) and as much as it lowers held-out copying. Their specificity against this control is -0.035 nats (random-control mean -0.023, $p = 0.567$), and the candidates selected under mean-ablation (Section 4.4) behave the same way (-0.138 nats, $p = 0.882$). On GPT-2, we therefore find no evidence under either baseline that the selected heads' effect is specific to in-context copying: they harm the format-matched task at least as much. A null contrast against one control task does not, however, rule out specificity relative to other tasks. DistilGPT2 differs. Its candidates lower the sequence task by only 0.113 nats, no more than random head sets (mean 0.107, $p = 0.280$), so their specificity is 0.472 nats ($p = 0.001$; mean-ablation candidates 0.382, $p = 0.001$). For DistilGPT2, the effect is larger on in-context copying than on a format-matched task that cannot be solved by copying. This is evidence of a task-dependent effect, not of an induction mechanism. The baseline match is also looser for DistilGPT2 (87.6% exact match on the sequence task vs. 90.0% on copying).

Sensitivity to the partition. We repeated selection and evaluation over 20 random 100/100 partitions of the 200 copying examples (Appendix E). On GPT-2, the same five candidates are selected in 19 of 20 partitions and exceed both head controls in all 20 ($p = 0.001$), with a median held-out effect of 0.441 nats (range 0.368–0.469). The arithmetic specificity contrast, however, gives $p \leq 0.05$ in 9 of 20 partitions (median $p = 0.054$), so whether it crosses the conventional threshold depends on the partition. On DistilGPT2, the same five heads are selected in all 20 partitions, and both head controls and the arithmetic contrast give $p = 0.001$ in each. The partitions overlap, so this is a sensitivity analysis rather than 20 independent replications.

4.4 RQ3: Does the replacement value change the conclusion?

Zero is a value a head's output may never take, so the RQ2 effect could reflect that distribution shift rather than the removal of a computation. After obtaining the primary results, we therefore reran the RQ2 pipeline with two replacement values that stay on the model's activation distribution (Section 3): the head's mean output (*mean-ablation*) and its output on another copying example (*resample-ablation*, specified in writing before it was run). Data, splits, selection rule, controls, seeds, and statistics are unchanged (Table 2).

The protocol still finds heads that beat both controls. Under either baseline and on both models, the reselected candidates exceed the random-head and layer-matched controls on held-out data (Monte Carlo $p = 0.001$ for each comparison). The matched-control protocol is therefore not tied to zero-ablation.

It finds different heads. On GPT-2, the mean-ablation candidates (Layer 10 Heads 11 and 0, Layer 7 Head 6, Layer 0 Heads 9 and 3) and the resample-ablation candidates (Layer 10 Heads 0, 11, 1, and 6, Layer 11 Head 2) share no head with the zero-ablation candidates, four of which lie in layers 0 and 1, but they share Layer 10 Heads 0 and 11 with each other. Per-head effects agree in the same way: on the discovery split, mean- and resample-ablation correlate at Pearson $r = 0.724$, whereas zero-ablation correlates with them at only 0.424 and 0.212.

The zero-selected heads lose their effect. In a post hoc comparison computed from the saved per-head effects, the five zero-selected GPT-2 heads have a held-out effect of 0.005 nats under mean-ablation instead of 0.412, below the mean of the mean-ablation random control (0.016). Because a pooled mean is itself a value a head may never produce, the pre-specified resample analysis tests the same heads with replacements the heads actually produce. Their effect is then 0.020 nats (CI $[0.006, 0.035]$), no larger than that of random head sets (mean 0.015, $p = 0.346$) or layer-matched sets ($p = 0.430$), and it is 0.005 and 0.011 nats with two other random choices of source examples. On GPT-2, the RQ2 effect of these heads is therefore consistent with replacing their output by an off-distribution value, not with removing information the copying task needs.

How much the baseline matters depends on the model. On DistilGPT2, resample-ablation selects exactly the mean-ablation candidates, which share one head with the zero-ablation set, and the zero-selected heads keep about a third of their effect (0.586 nats under zero, 0.182 under mean, 0.201 under resample). Under resample that remainder exceeds layer-matched head sets ($p = 0.001$) but not clearly random ones ($p = 0.075$). On Qwen2.5-1.5B-Instruct, mean-ablation reselects the same five heads as zero-ablation, and they keep most of their effect (0.208 to 0.155 nats; Appendix C); we did not run resample-ablation on Qwen2.5.

Task specificity does not return. Under mean-ablation, the GPT-2 candidates' contrast against arithmetic gives $p = 0.014$ (0.168 nats versus a random-control mean of 0.021), but against the format-matched sequence task the same candidates show no specificity ($p = 0.882$; Section 4.3). We therefore do not read the arithmetic result as evidence of task specificity.

4.5 RQ3 on an established circuit task: indirect object identification

To test whether baseline dependence also arises on a natural-language circuit task, we ran indirect object identification (IOI; 4) on GPT-2 small: 200 prompts from the 15 templates of Wang et al. [4] with single-token names, places, and objects, a 100/100 discovery/held-out split, and the logit difference between the indirect object and the subject as the metric. GPT-2 prefers the indirect object on 97% (discovery) and 100% (held-out) of prompts. We compared three replacement values at the same pre-projection location: zero, the discovery-split mean, and a *resample* value, the head's activation on the "ABC" variant (the second subject replaced by a third name) of a different prompt with the same template. Selection, controls, and statistics are those of RQ2.

Under every baseline the selected heads exceed both matched controls ($p = 0.001$ each). The baselines disagree on which heads matter: zero shares only 2 of its top-5 heads with mean or with resample, whereas mean and resample share 4 of 5 (Spearman ρ over all 144 heads: zero vs. mean 0.74, zero vs. resample 0.47, mean vs. resample 0.65). Zero-ablation selects three layer-0/1 heads that neither on-distribution baseline selects, and its held-out effect is 68% larger (1.55 vs. 0.92 and 0.93). Under every baseline the top-5 includes the S-inhibition heads 8.6 and 8.10 reported by Wang et al. [4], and their head 7.9 is in the top-5 under mean and resample (Appendix D).

Table 2: Robustness to the replacement value ($N = 200$ per task; Δ in nats). The first block compares per-head effects under zero and mean ablation. The second repeats RQ2 with candidates reselected under mean-ablation. The third evaluates the zero-selected candidates under zero and mean ablation and was computed post hoc from the saved per-head effects. The last block is the pre-specified resample-ablation analysis (discovery $N = 99$, held-out $N = 100$): held-out Δ of the stated heads, p-values against random and layer-matched head sets, and the per-head Pearson r with the other two baselines on the discovery split. Intervals are 95% percentile-bootstrap CIs over examples; †5th–95th percentile of 1,000 control draws.

	GPT-2 small		DistilGPT2	
Quantity	Value	Interval	Value	Interval
Zero vs. mean per-head Δ: Pearson r	0.424	–	0.553	–
Zero vs. mean per-head Δ: Spearman ρ	0.495	–	0.871	–
Top-5 overlap, mean- vs. zero-selected	0/5	–	1/5	–
Candidate Δ, held-out	0.176	[0.157, 0.194]	0.445	[0.388, 0.508]
Random-control Δ, held-out	0.016	$[-0.034, 0.057]^{\dagger}$	0.060	$[-0.012, 0.172]^{\dagger}$
Layer-matched Δ, held-out	0.009	$[-0.071, 0.058]^{\dagger}$	0.030	$[-0.024, 0.067]^{\dagger}$
Rank stability ρ	0.948	–	0.960	–
p: random / layer-matched / specificity	0.001 / 0.001 / 0.014		0.001 / 0.001 / 0.001	
Zero-selected heads, held-out Δ under zero	0.412	–	0.586	–
Zero-selected heads, held-out Δ under mean	0.005	–	0.182	–
Zero-selected heads under resample	0.020	[0.006, 0.035]	0.201	[0.173, 0.232]
p: random / layer-matched		0.346 / 0.430		0.075 / 0.001
Resample-selected, held-out Δ	0.179	[0.159, 0.200]	0.586	[0.516, 0.659]
p: random / layer-matched		0.001 / 0.001		0.001 / 0.001
Overlap with zero / mean top-5		0/5 / 2/5		1/5 / 5/5
Pearson r with zero / mean		0.212 / 0.724		0.473 / 0.946

5 Discussion and limitations

Scope of the causal claim. We do not claim these five heads "compute induction" in a mechanistic sense. Using the causal-claim ladder (correlation $<$ decodability $<$ intervention effect $<$ specific causal contribution $<$ mechanistic role), our evidence supports a held-out-replicated intervention effect that exceeds matched controls (the third rung), but only relative to a stated replacement value. The zero-selected GPT-2 heads reach this rung under zero-ablation and not under mean- or resample-ablation. The Qwen2.5 heads reach it under both. No GPT-2 specificity contrast supports the fourth rung. A mechanistic account would require further evidence, such as activation patching, input-dependence analysis, or directional interventions.

Task-control mismatch. The arithmetic control differs from copying in template, computational demands, tokenization, and baseline performance, and it sits at an accuracy floor. A difference in ablation effects between the two tasks therefore cannot be attributed uniquely to a copying-specific computation, and we treat the arithmetic contrast as a limited diagnostic. The sequence-continuation control (Section 4.3) matches the template, answer length, and baseline. It removes the GPT-2 contrast but not the DistilGPT2 one, and it is still only one alternative rule.

Ablation baseline. We tested zero, mean, and resample replacement (resample on copying for GPT-2 and DistilGPT2, and on IOI). We did not test noise injection or optimal ablation [5], or resample-ablation on Qwen2.5. Because the baseline changed the selected heads on GPT-2 and IOI but barely on Qwen2.5, whether it matters has to be checked for each model and task.

Model scope. DistilGPT2 is distilled from GPT-2, so it is not an independent replication. On it, the naive and correct effects are again near-uncorrelated ($r = 0.001$, 0/5 overlap), the arithmetic floor recurs, and the selected heads exceed both matched controls (Appendix B). Copying accuracy is lower (90.0%), so the near-ceiling example is specific to GPT-2. Qwen2.5-1.5B-Instruct is a different architecture family and reproduces the projection confound and the matched-control separation (Appendix C). We did not test models larger than 1.5B parameters, and the new partition, sequence-control, and copying resample analyses were run on GPT-2 and DistilGPT2 only, because one Qwen pipeline run took 3.3 CPU-hours.

Tasks, partitions, and sample size. IOI is the only established circuit task we tested. We did not test others, such as Greater-Than [13]. The partition analysis reuses the same 200 copying examples and uses zero-ablation only. The sequence control has 129 examples, 92 of them numerals.

Broader implication. Each check changed a conclusion: the naive hook selected disjoint heads; accuracy hid information at the floor and near the ceiling; the specificity contrast blocked a task-specificity claim that the larger GPT-2 run does not support; and the replacement value changed which heads were selected and, for GPT-2, removed the selected heads' effect under both on-distribution baselines. A single-head ablation report should therefore state the intervention point with a test that it isolates one head, use a continuous metric beside accuracy, select on held-out-separated data against matched controls, and report the replacement value together with at least one on-distribution baseline.

6 Conclusion

A head ablation is not self-certifying evidence of a specific causal role. Hook placement can ablate the wrong quantity, accuracy is censored at a floor and coarse near a ceiling, and held-out matched controls separate a replicating effect from arbitrary disruption. That effect can still depend on the replacement value, and task specificity can still fail to replicate.

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A Supplementary figure: RQ0 per-head comparison

Figure 2 plots the per-head effect (correct pre-projection ablation vs. naive post-projection ablation) for all 144 heads on the controlled copying-task discovery split, corresponding to the summary statistics in Section 4.1.

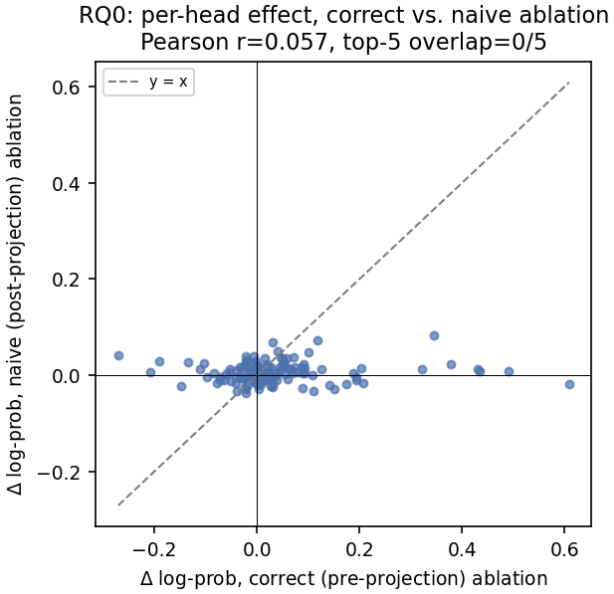


Figure 2: Per-head Δ log-prob under the correct (pre-projection) vs. naive (post-projection) ablation method, all 144 heads, controlled copying-task discovery split. A valid replication of the naive method would lie near the $y = x$ line. Instead the two methods are nearly uncorrelated (Pearson $r = 0.057$).

B Replication on a second model (DistilGPT2)

As a first robustness check (Section 5), we reran the identical pipeline (Section 3), unchanged except for the model, on DistilGPT2 (6 layers, 12 heads/layer, 72 heads total). All data, metrics, ablation methods, splits, and control procedures are exactly as in Section 3, and only the model differs. Results are in Table 3.

The candidate heads found on DistilGPT2 (Layer 0 Heads 1, 7, 9; Layer 3 Head 2; Layer 5 Head 0) are, as expected, different indices from GPT-2 small's. The two models are different parameter sets, so this is not evidence the same head performs the same role across models, only that the same *protocol* (discovery/held-out selection against matched controls) again separates a real, held-out-replicated effect from the control distributions. We report this as a second data point for the paper's methodological claim, not as evidence about a shared or universal "induction circuit" across GPT-2 and DistilGPT2. DistilGPT2 is distilled from GPT-2, so it is not an independent model. Its arithmetic specificity contrast ($p = 0.001$) comes from a single run with no larger rerun and is subject to the same task mismatch as GPT-2's (Section 5), so we do not treat it as evidence of task specificity.

C Replication on a different architecture (Qwen2.5-1.5B-Instruct)

We reran the identical pipeline, with both zero- and mean-ablation and unchanged except for the model, on Qwen2.5-1.5B-Instruct [10]. Its pre-projection hook sits on the input of `o_proj`. Executable tests on a small random Qwen2 configuration assert the same intervention semantics as for GPT-2. The run took 11,976 CPU-seconds with 9.0 GiB peak resident memory and was performed once (Table 4). The projection confound recurs ($r = 0.108$, 0/5 overlap). The zero-ablation candidates (Layer 8 Head 3, Layer 14 Head 3, Layer 18 Head 8, Layer 13 Head 4, Layer 2 Head 3) exceed both

Table 3: DistilGPT2 replication of Table 1, with the same procedure and Ns. The intervention-semantic and matched-control findings replicate, and the arithmetic floor recurs. Copying accuracy is lower (90.0%), so GPT-2's near-ceiling example does not replicate. The specificity comparison gives Monte Carlo $p = 0.001$.

Quantity	Value	Interval	N / draws
Copying-task accuracy	0.900	[0.855, 0.940]	200
Arithmetic accuracy (floor)	0.000	[0.000, 0.000]	200
Copying-task log-prob (nats)	-0.96	$[-1.07, -0.85]$	200
Arithmetic log-prob (nats)	-7.38	$[-7.50, -7.25]$	200
RQ0: Pearson r(naive, correct)	0.001	–	72 heads
RQ0: top-5 head overlap	0/5	–	–
Candidate Δ, held-out	0.586	[0.511, 0.661]	100
Random-control Δ, held-out	0.080	$[-0.026, 0.231]^{\dagger}$	1000
Layer-matched Δ, held-out	0.054	$[-0.017, 0.128]^{\dagger}$	1000
Rank stability (ρ, disc. vs. held-out)	0.957	–	72 heads
Candidate specificity	0.492	–	–
Random-control specificity	0.023	$[-0.159, 0.222]^{\dagger}$	1000

Monte Carlo p(candidate $\geq$ random control, held-out Δ) $= 0.001$; p(candidate $\geq$ layer-matched, held-out Δ) $= 0.001$; p(candidate $\geq$ random control, specificity) $= 0.001$. †5th–95th percentile.

controls, and under mean-ablation the same five heads are selected and still exceed both controls. The specificity contrast against arithmetic is significant under both baselines ($p = 0.004$ and 0.007), subject to the task-mismatch caveat. Exact match on arithmetic is 0.0%, but 23.0% when the answer is written without its leading space (Section 4.2).

Table 4: Qwen2.5-1.5B-Instruct ($N = 200$ per task, 336 heads). Intervals are 95% percentile-bootstrap CIs over examples; †5th–95th percentile of 1,000 control draws.

	Zero-ablation		Mean-ablation	
Quantity	Value	Interval	Value	Interval
Copying-task accuracy	0.990	[0.975, 1.000]		
Arithmetic accuracy	0.000	[0.000, 0.000]		
Copying-task log-prob (nats)	-0.38	$[-0.43, -0.34]$		
Arithmetic log-prob (nats)	-5.31	$[-5.44, -5.17]$		
RQ0: r / ρ (naive vs. correct)	0.108 / 0.176	–		
RQ0: top-5 head overlap	0/5	–		
Zero vs. mean Δ: r / ρ			0.925 / 0.756	–
Top-5 overlap with zero-selected			5/5	–
Candidate Δ, held-out	0.208	[0.176, 0.243]	0.155	[0.129, 0.182]
Random-control Δ, held-out	0.012	$[-0.010, 0.046]^{\dagger}$	0.006	$[-0.007, 0.031]^{\dagger}$
Layer-matched Δ, held-out	0.004	$[-0.012, 0.019]^{\dagger}$	0.003	$[-0.008, 0.015]^{\dagger}$
Rank stability ρ	0.927	–	0.850	–
Candidate specificity	0.376	–	0.349	–
Random-control specificity	-0.007	$[-0.223, 0.205]^{\dagger}$	0.001	$[-0.203, 0.189]^{\dagger}$
p: random / layer / specificity	0.001 / 0.001 / 0.004		0.001 / 0.001 / 0.007	

D Indirect object identification details

Table 5 gives the selected heads and the ranks of the heads that Wang et al. [4] identify, under each replacement value (Section 4.5). The resample source for each prompt is drawn uniformly (fixed seed) from the other prompts of the same template within the same split, so source and target have identical token length and every position plays the same syntactic role. Name mover 9.6 ranks 141st or lower under all three baselines, so ablating it alone increases the logit difference. Wang et al. [4] report backup heads that compensate when name movers are removed. Our single-head result is consistent with such compensation, although we did not test it. The run is saved in `results/naacl_ioi/gpt2_ioi_ablation_methods.json`.

Table 5: IOI on GPT-2 small (200 prompts, 100/100 discovery/held-out). Head $\ell.h$ is layer ℓ, head h; top-5 heads in rank order. Held-out effect: mean drop in logit difference (IO minus subject) for the five selected heads, with 95% bootstrap CI. Ranks are by mean effect over all 200 prompts; rank 144 is the most negative effect.

	Zero	Mean	Resample
Top-5 heads	0.10, 1.3, 8.6, 8.10, 0.7	8.6, 5.5, 8.10, 7.9, 5.9	8.6, 8.10, 5.5, 7.9, 7.3
Held-out effect	1.55 [1.42, 1.69]	0.92 [0.84, 1.01]	0.93 [0.84, 1.02]
p vs. random / layer-matched	.001 / .001	.001 / .001	.001 / .001
Rank stability ρ	0.94	0.97	0.91
Rank (of 144) of heads reported by Wang et al. [4]			
S-inhibition 7.3 / 7.9 / 8.6 / 8.10	14/6/3/5	7/4/1/2	5/4/1/2
Name mover 9.6 / 9.9 / 10.0	141/26/15	141/16/9	142/12/9
Negative name mover 10.7 / 11.10	144/143	144/143	144/143

E Partition sensitivity

Table 6 summarizes the 20 random 100/100 partitions of the 200 copying examples (permutation seeds 1000–1019). For each partition we reselect the top-5 heads on its discovery half and repeat the RQ2 statistics on its held-out half, using the zero-ablation per-head effects computed once on all 200 examples. Restricted to the original partition, these effects equal the saved per-head values exactly. Results are in `<model>_camera_ready_robustness.json`.

Table 6: Partition sensitivity over 20 random 100/100 discovery/held-out partitions (zero-ablation). Ranges are minimum–maximum over partitions. Overlap is with the candidates of the original partition.

	GPT-2 small	DistilGPT2
Partitions with the original top-5 (5/5 overlap)	19/20	20/20
Candidate held-out Δ, median [range]	0.441 [0.368, 0.469]	0.578 [0.535, 0.643]
Rank stability ρ, median [range]	0.966 [0.937, 0.979]	0.961 [0.923, 0.981]
Partitions with $p \leq 0.05$ vs. random / layer-matched	20/20 / 20/20	20/20 / 20/20
Arithmetic specificity, median [range]	0.168 [-0.014, 0.196]	0.484 [0.441, 0.549]
Partitions with arithmetic specificity $p \leq 0.05$	9/20	20/20
Arithmetic specificity p, median [range]	0.054 [0.039, 0.650]	0.001 [0.001, 0.001]

F Reproducibility details

All results are produced by `scripts/run_neurips_experiments.py` and saved to `neurips/experiment_results/gpt2_results.json` (main results, Table 1) and `distilgpt2_results.json` (replication, Table 3). These files record the device, torch (2.8.0) and transformers (4.57.3) versions, all random seeds, per-split example counts, the number of bootstrap resamples (2,000) and control draws (1,000), the full per-head Δ vector for every split and method, and per-example log-probabilities for every baseline. Exact model revisions and hashes are recorded in `neurips/experiment_results/reproducibility_provenance.md`. A separate one-time, higher-power rerun of the main GPT-2 pipeline with $N{=}500$ (rather than 200) per task, run once and reported regardless of outcome, is saved to `gpt2_n500_results.json`. Its preserved held-out effect and weaker specificity contrast are reported in Section 4.3. Figures are produced by `scripts/make_figures.py`, which reads only the primary JSON file and performs no additional computation. A claim-to-evidence map (`neurips/claim_evidence_map.md`) lists, for every quantitative statement in this paper, the exact JSON field it was read from. Every file named in this appendix, together with the pipeline source, the figure script, and a README giving the pinned environment and the exact command for each reported run, is available at `https://github.com/TheClassicTechno/InterpretabilityNLPProject`. No dataset is shipped: all task families are generated in memory by `src/dataset.py` from the seeds recorded in each result file (the sequence-continuation task is deterministic).

The analyses added for the camera-ready version are saved separately and leave the original result files unchanged: `{gpt2,distilgpt2}_meanablation_results.json` and `mean_vs_zero_posthoc.json` (Section 4.4), `qwen2.5-1.5b-instruct_meanablation_results.json` and `arithmetic_format_diagnostic.json` (Appendix C), `results/naacl_ioi/` (Section 4.5), and `{gpt2,distilgpt2}_camera_ready_robustness.json` (Section 4.3, Appendix E), and `{gpt2,distilgpt2}_resample_copying.json` (Section 4.4). The design of the partition, sequence-control, and resample analyses was fixed in `neurips/camera_ready_design.md` before they were run.

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   Justification: No model is trained. Section 3 gives the split sizes and procedure, fixed top-$k = 5$ candidate count, 2,000 bootstrap resamples, and 1,000 control draws. These settings were fixed before results were inspected; the partition, sequence-control, and resample analyses added for the camera-ready version were specified in writing before they were run. Each result file in the public repository records the full configuration used to produce it, including seeds, per-split example counts, and task parameters.

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   Justification: Section 3 reports one 10-core Apple M5 CPU, 24 GB unified memory, no GPU, and about 1.5 GB of local storage. Measured wall times and peak resident memory are 198.99 s and 1.94 GiB for the primary GPT-2 pipeline, 57.62 s and 1.90 GiB for DistilGPT2, 428.56 s and 3.22 GiB for GPT-2 at $N = 500$, and 301.43 s (1.90 GiB) and 82.08 s (1.63 GiB) for the camera-ready partition and sequence-control analyses on GPT-2 and DistilGPT2, and 914.99 s (2.14 GiB) and 325.98 s (2.10 GiB) for the copying resample analysis. The Qwen2.5-1.5B-Instruct pipeline used 11,976 CPU-seconds and 9.0 GiB (Appendix C). The provenance record lists every run with its command and measurements.

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